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Non-Colliding Biometric Identities for Digital Entities: Geometry, Capacity, and Million-Scale Virtual Identity Provisioning

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Abstract

Digital entities such as AI agents and humanoid robots increasingly operate alongside real humans, yet their identity infrastructure is based on credentials rather than embodied biometric identity. We introduce Biometric Identity Provisioning (BIP), a new problem and solution framework that addresses: given an enrollment gallery of real human identities, provision virtual identities that are non-colliding with every enrolled identity, maintain sufficient inter-class separability, and are realizable as high-fidelity face images. The key geometric insight is that real face identities occupy a low-dimensional subspace of the embedding hypersphere, leaving no residual subspace for virtual identities. Hence, virtual identities must instead be allocated as unclaimed gaps within the real face manifold itself. BIP is therefore a constrained packing problem: available gaps vastly exceed any foreseeable enrollment scale, and provisioned identities remain non-colliding even as new real identities are subsequently enrolled. Grounded in this geometry, our repulsion-based allocation is not bounded by any fixed provisioning count; we demonstrate 10M non-colliding virtual identity embeddings against a gallery of 360K real identities. Realizing these embeddings as face images requires a generator that operates outside the training distribution of real face images; we introduce GapGen, a gap-aware generator trained with a curriculum that progressively extends synthesis into non-colliding regions, validated at 1M photorealistic virtual face images. We further construct v-LFW, a virtual counterpart to LFW face dataset, with protocols for virtual face verification, cross-reality matching, real-vs-virtual detection, and unified recognition and detection.

1 Introduction

Identity systems are being reshaped by a new class of actors: AI agents autonomously coordinate enterprise workflows [30], humanoid robots interact face-to-face with people in service environments, and digital co-workers operate under zero-trust policies alongside human employees. This is not a projected future: a recent industry survey reports that 40% of organizations already have AI agents in production, with another 31% running pilots or tests [5]. Microsoft has launched dedicated infrastructure to govern these agents as "first-class" identities [23]. Yet this infrastructure assigns credential identity (tokens, certificates, access scopes), not *embodied biometric identity*. As digital entities operate in physical spaces alongside real humans, a question arises: *how do we provision digital entities with persistent, unique biometric identities that do not collide with any enrolled real human identity?* We term this the **Biometric Identity Provisioning (BIP)** problem (see Fig. 1).

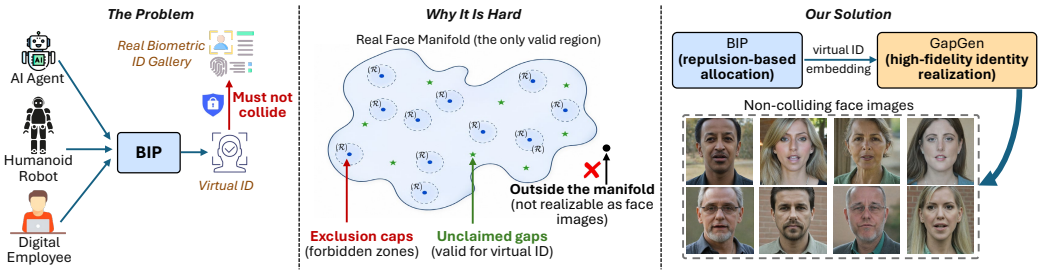


Figure 1: Should a digital entity have a distinctive face? As AI agents, humanoid robots, and digital employees operate alongside real humans, they require persistent biometric identities that must not be mistaken for any enrolled real person (left). This is geometrically non-trivial: real face identities occupy a low-dimensional manifold with no residual subspace, forcing virtual identities into the narrow unclaimed gaps between exclusion caps; outside the manifold, embeddings cannot be realized as natural face images (middle). BIP provisions non-colliding virtual identity embeddings via repulsion-based allocation, and GapGen realizes them as high-fidelity 1024×1024 portrait faces (right).

Face is the natural entry point for biometric identity provisioning for virtual entities: it is the modality most directly involved in human-entity interaction, the most mature in generative modeling, and the one for which large-scale enrollment galleries already exist. A substantial body of work has studied synthetic face generation, but with a fundamentally different objective: producing training data for face recognition models. Methods such as Arc2Face [24] and Vec2Face/Vec2Face+ [37, 36] condition face generation on real human identity embeddings, generating faces that correspond to existing real people in embedding space; they are *identity cloners*, not *identity creators*. Methods without this conditioning, such as DCFace [16], optimize for separability among synthetic faces, but whether generated identities collide with real enrolled humans is simply not a design concern; empirically, we show such collisions do occur at non-trivial rates. Thus, a synthetic identity useful for training a classifier is not necessarily a biometric identity suitable to assign to a digital entity. Existing methods neither define nor jointly satisfy the requirements of high-fidelity realization, inter-class separability, and guaranteed **non-collision** with any enrolled real humans.

Hence, we propose a new formulation, termed BIP: synthetic identities shall be treated not merely as generated images, but as allocatable positions in a shared biometric identity space. We define BIP as:

> *Given a mapping from face images to d-dimensional L2-normalized embeddings on $\mathbb{S}^{d-1}$ and an enrollment gallery $\mathcal{R} \subset \mathbb{S}^{d-1}$, provision a set of virtual embeddings $\mathcal{V} \subset \mathbb{S}^{d-1}$ satisfying non-collision with $\mathcal{R}$ at threshold τ, inter-class separability within $\mathcal{V}$, and high-fidelity realizability.*

A natural but naïve geometric view treats BIP as a **capacity** problem: the number of separable positions on $\mathbb{S}^{d-1}$ at threshold τ is arbitrarily large even within the effective face subspace, suggesting ample room for virtual identities. This view is misleading: ***(1) No residual subspace exists.*** Virtual identities satisfying BIP constraints do not occupy arbitrary low-energy directions of the hypersphere; their PCA energy distribution closely follows that of real identities, with both concentrating over 95% of their variance within the same $k{=}269$ leading principal components out of $d{=}512$. Valid virtual identities must occupy *unclaimed gaps within the real face manifold* rather than a geometrically separate region. ***(2) Capacity does not equal realizability.*** Embedding positions or simply embeddings satisfying the non-collision constraint are not automatically realizable as high-fidelity face images: the further a virtual identity is pushed from real identity territories, the harder it becomes for a generative model to generate it as a natural face. BIP is, therefore, a constrained identity-allocation problem on the real identity manifold, not an unconstrained sampling problem on $\mathbb{S}^{d-1}$.

Grounded in this geometric view, we address BIP through an identity-first pipeline that separates allocation in embedding space from realization in image space:

(1) Formalization. We introduce the first formal definition of BIP and introduce three evaluation metrics operationalizing its requirements: non-collision, inter-class separability, and perceptual image quality. ***(2) Geometry and allocation.*** We characterize the geometry of the real identity manifold and show that safe virtual identities must occupy directional gaps within it. We derive the **repulsion direction**, z^* as a proposed heuristic that moves candidates away from real identity clusters, enriched with PCA-aware noise to maintain face-manifold compatibility. Combined with exact hard checks, the allocation strategy is not bounded by any fixed provisioning count: the face manifold contains vastly

more unclaimed gaps than any foreseeable enrollment gallery, and we realize and demonstrate 1M non-colliding virtual identity embeddings against a gallery of 360K real identities with no observed collision. ***(3) Realization via GapGen.*** We introduce a gap-aware generator trained with a progressive curriculum that extends face synthesis into the non-collision region. ***(4) Benchmark.*** We construct **v-LFW**, a virtual counterpart to LFW face dataset with 5,749 provisioned identities and 13,233 images, and introduce protocols spanning virtual face verification, cross-reality matching, real-vs-virtual detection, and unified recognition and detection. To support these protocols, we provide IAPCT (Identity-Anchored Patch Consistency Transformer) as a lightweight diagnostic evaluation tool.

2 Related Work

Synthetic Face Generation for Recognition. Synthetic face data has been widely explored as a privacy-preserving alternative to web-collected face datasets. Early work such as SynFace [26] and SFace [4] uses class-conditional generative models, while DigiFace-1M [2] adopts graphics-based rendering. More recent methods improve identity diversity via diffusion: IDiff-Face [3] introduces identity-conditioned diffusion, and DCFace [16] disentangles identity and style via dual-condition diffusion. Arc2Face [24] and Vec2Face and Vec2Face+ [37, 36] further generate identity- and attribute-controllable face datasets from recognition-based features, with Vec2Face+ explicitly controlling inter-class separability, intra-class variation, and identity consistency. VIGFace [17] is especially related because it pre-assigns virtual identities in feature space before synthesis. However, these methods are designed primarily for privacy-friendly or high-performing face recognition training data. They do not formulate identity generation as gallery-conditioned provisioning: whether a generated identity collides with an enrolled real human identity is not a primary constraint, and persistent assignment to digital entities is outside their problem setting.

Hyperspherical Identity Allocation. Modern face recognition systems map faces to normalized embeddings on a high-dim hypersphere, where identity verification uses cosine similarity or angular distance. SphereFace [22], CosFace [33], ArcFace [7], and AdaFace [15] introduce angular margin losses that structure this hypersphere to improve inter-class separation and intra-class compactness, with AdaFace further adapting the margin to image quality via feature norms. The most closely related work to BIP is HyperFace [28], which formulates synthetic face data generation as a packing problem on the recognition hypersphere and optimizes identity embeddings via gradient descent, while regularizing embeddings on the face manifold. However, HyperFace maximizes inter-class separation *among synthetics only*; non-collision with real humans is not a design objective. BIP differs fundamentally: the enrollment gallery $\mathcal{R}$ is a hard constraint, non-collision with every identity in $\mathcal{R}$ at threshold τ is the primary requirement, and our geometric analysis shows that valid virtual identities must reside within the real identity manifold rather than a geometrically separate space.

Identity-Aware Forensics. Deepfake and face forgery detection has been formulated as binary classification between real and manipulated images. Many detectors exploit low-level artifacts, frequency-domain cues, blending boundaries, or generator-specific traces [27, 29, 9, 25], but they often generalize poorly across manipulation types and generation models. Identity-aware forensics treats identity consistency as a forensic signal: ID-Reveal [6] learns person-specific motion patterns and detects manipulations as deviations from expected behavior, while other works exploit identity inconsistency or face recognition features to detect face swaps [38, 18]. These methods assume that real identities already exist and ask whether a given image is manipulated. BIP addresses a complementary question: how to provision new biometric identities guaranteed to lie outside all real identity territories. The resulting non-overlapping partition of the embedding space between real and virtual identities is a direct benefit of the BIP constraints, not a product of forensic training.

3 Methodology

3.1 Formalization

Let $\phi : \mathcal{X} \to \mathbb{S}^{d-1}$ denote a face recognition encoder that maps a face image x to a d-dimensional L2-normalized embedding on the unit hypersphere $\mathbb{S}^{d-1}$.

Assumption 1 (Angular-margin Encoder)**.** *The encoder ϕ is trained with an angular margin loss (*e.g.*, ArcFace [7] or AdaFace [15]). Embeddings with $\cos(\phi(x), \phi(x')) \geq \tau$ are accepted as the same identity, where $\tau \in (0, 1)$ is set at the operating point of the recognition system (Refer to*

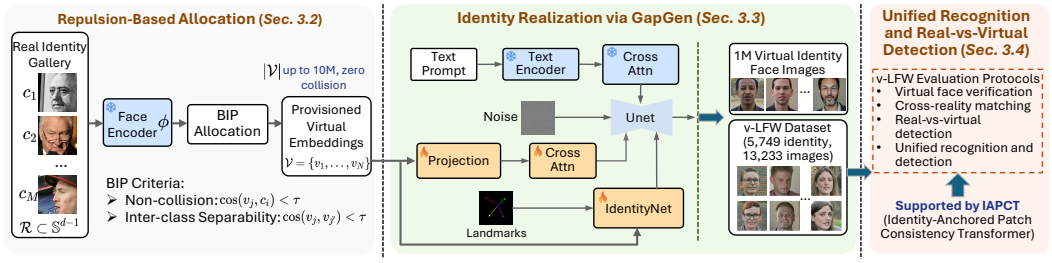


Figure 2: BIP pipeline. Left: Repulsion-based allocation provisions $\mathcal{V}=\{v_1, \ldots, v_N\}$ satisfying $\cos(v_j, c_i)<\tau$ and $\cos(v_j, v_{j'})<\tau$, scaling to $|\mathcal{V}|$=10M with zero observed collision. Middle: GapGen renders each $s \in \mathcal{V}$ into a 1024×1024 face image $\tilde{x}=G(s)$, producing 1M virtual identity images and the v-LFW benchmark. Right: v-LFW supports four protocols spanning virtual face verification, cross-reality matching, real-vs-virtual detection, and unified recognition and detection, supported by IAPCT as a lightweight diagnostic tool.

Appx. B for details on τ). Under Assumption 1, τ defines an angular boundary on $\mathbb{S}^{d-1}$: any two embeddings within angular distance $\arccos(\tau)$ are recognized as the same identity.

Each real identity $i \in \{1, \ldots, M\}$ with images $\mathcal{X}_i = \{x_i^{(1)}, \ldots, x_i^{(n_i)}\}$ is represented by the dominant direction of its embedding cluster:

$$c_i = \text{normalize}\left(\sum_{k=1}^{n_i} \phi\left(x_i^{(k)}\right)\right), \tag{1}$$

with $c_i = \phi(x_i^{(1)})$ when $n_i = 1$. The enrollment gallery is $\mathcal{R} = \{c_1, \ldots, c_M\} \subset \mathbb{S}^{d-1}$.

Definition 1 (Biometric Identity Provisioning)**.** *Given an encoder ϕ satisfying Assumption 1, an enrollment gallery $\mathcal{R} \subset \mathbb{S}^{d-1}$ of M real identity centroids, a verification threshold τ, and a provisioning count N, the **Biometric Identity Provisioning (BIP)** problem is to generate $\mathcal{V} = \{v_1, \ldots, v_N\} \subset \mathbb{S}^{d-1}$ satisfying:*

$$\cos(v_j, c_i) < \tau, \quad \forall i \in \{1, \ldots, M\},\ \forall j \in \{1, \ldots, N\} \tag{2}$$

$$\cos(v_j, v_{j'}) < \tau, \quad \forall j \neq j'. \tag{3}$$

The realizability requirement that each v_j corresponds to a high-fidelity face image whose re-encoded embedding satisfies Eqs. (2)–(3), is addressed by the gap-aware generator in Sec. 3.3. Fig. 2 summarizes the full pipeline of allocation, realization, and evaluation.

Remark 1. *The constraints in Definition 1 are* gallery-relative*: they are guaranteed w.r.t. $\mathcal{R}$ at provisioning time. When new real identities $\Delta\mathcal{R}$ are subsequently enrolled, any v_j with $\cos(v_j, r') \geq \tau$ for some $r' \in \Delta\mathcal{R}$ must be revoked and reassigned against the updated gallery $\mathcal{R} \cup \Delta\mathcal{R}$. This requires $O(|\mathcal{V}| \times |\Delta\mathcal{R}|)$ cosine operations in the exact case, reducible with approximate nearest-neighbor indexing [13]. For open-world robustness against subsequently enrolled real identities, a safety buffer can be applied at provisioning time; Appx. C (Proposition 2) formalizes this guarantee, and open-world generalization is empirically validated in Sec. 4.*

BIP Criteria. Three metrics directly operationalize Definition 1. ***(i) Non-collision:*** the percentage of provisioned virtual identities $v_j \in \mathcal{V}$ satisfying Eq. (2) against all M enrolled real centroids in $\mathcal{R}$; a score below 100% means some virtual identities are indistinguishable from real humans in the recognition system, posing a direct privacy risk. ***(ii) Inter-class separability (Inter-Sep):*** the percentage of virtual identity pairs $(v_j, v_{j'})$ satisfying Eq. (3); a score below 100% means some virtual entities share an identity, making them unreliable as distinct assignable identities. ***(iii) FID*** [10] (against FFHQ [14]) measures perceptual image quality of generated virtual identities against portrait photographs. Unlike prior synthetic face methods that produce low-resolution 112×112 training crops, digital entities require portrait-quality images displayable and recognizable in physical environments.

3.2 Geometry of the Real Identity Manifold and Repulsion-Based Allocation

Geometry of the Real Identity Manifold. ArcFace ResNet-100 [7] trained on Glint360K [1] is used with ϕ (d=512), and all M=360,232 identity centroids forming $\mathcal{R}$. To characterize where virtual identities can be validly placed and to guide provisioning diversity, we perform PCA on these

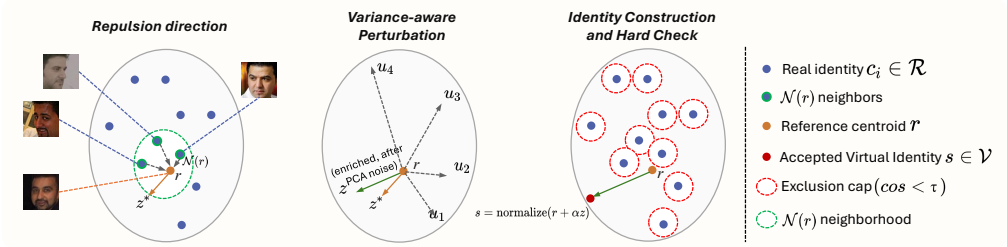


Figure 3: Repulsion-based virtual identity allocation. Left: $z^*{=}{-}m/\|m\|_2$ points away from the weighted centroid of nearest real neighbors $\mathcal{N}(r)$. Middle: z^* is enriched with PCA-aware noise along $\{u_k\}$ (weighted by σ_k) to obtain z. Right: Candidate $s{=}\text{normalize}(r{+}\alpha z)$ is accepted only if $\cos(s, c_i){<}\tau$ for all $c_i \in \mathcal{R}$ and $\cos(s, v_j){<}\tau$ for all $v_j \in \mathcal{V}_t$; failed candidates are resampled.

centroids in the ambient Euclidean space, yielding eigenvectors $U = [u_1, \ldots, u_d]$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$, where $k \in \{1, \ldots, d\}$ denotes rank in explained variance, and per-direction standard deviations $\sigma_k = \sqrt{\lambda_k}$; both $\{u_k\}$ and $\{\sigma_k\}$ are reused in the allocation strategy below.

Observation 1 (Low-dimensional Real Identity Manifold)**.** *Let the* principal energy *for dimension k be $E(k) = \left(\sum_{k'=1}^{k} \lambda_{k'}\right) / \left(\sum_{k'=1}^{d} \lambda_{k'}\right)$, where $k' \in \{1, \ldots, d\}$ is a summation index. The real identity centroids in $\mathcal{R}$ concentrate over 95% of their total variance within the top $k{=}269$ principal components (Fig. 7 in Appx. C.1), despite residing in $\mathbb{S}^{511}$. Since ϕ is trained on real face images, realizable face embeddings are expected to remain concentrated near the dominant real-face subspace. Low-variance residual directions may exist in the ambient embedding space, but they are unlikely to provide a reliable region for high-fidelity face realization. Valid virtual identities should therefore occupy* unclaimed gaps within the real face manifold*, rather than a geometrically separate region.*

Observation 1 rules out orthogonal or free-subspace constructions [17] and motivates the following gap-allocation strategy; the theoretical capacity of the face manifold under this constraint is derived in Appx. C (Proposition 1).

Repulsion Direction. Given a reference centroid $r \in \mathcal{R}$, let $\mathcal{N}(r) = \{c_{n_1}, \ldots, c_{n_K}\}$ be its K nearest neighbors in $\mathcal{R}$, with softmax weights:

$$w_{n_k} = \frac{\exp(-d(r,\, c_{n_k})/t)}{\sum_{k'=1}^{K} \exp(-d(r,\, c_{n_{k'}})/t)}, \tag{4}$$

where $d(r, c) = 1 - \cos(r, c)$ and t denotes temperature.

Remark 2 (Repulsion Direction)**.** *Define z^* as the negation of the weighted neighborhood centroid:*

$$z^* = -\frac{\sum_{k=1}^{K} w_{n_k}\, c_{n_k}}{\left\|\sum_{k=1}^{K} w_{n_k}\, c_{n_k}\right\|_2}. \tag{5}$$

z^* serves as a repulsive heuristic*: it moves candidates away from the weighted neighborhood centroid of r, improving the probability of a candidate passing the checks of Eqs. (9)–(10). Non-collision is guaranteed exclusively by the checks; see Appx. C for an analysis of z^* after spherical normalization.*

To improve diversity while keeping provisioned identities on the real face manifold, we enrich z^* with *PCA-aware noise*, random perturbations aligned to the principal directions $\{u_k\}$:

$$z = \text{normalize}\left(z^* + \kappa \sum_{k=1}^{d} \eta_k \sigma_k u_k\right), \quad \eta_k \sim \mathcal{N}(0, 1), \tag{6}$$

where $\kappa \geq 0$ balances direction fidelity against sample diversity. Weighting by σ_k ensures perturbations are larger along high-variance principal directions and smaller along low-variance ones, keeping provisioned embeddings aligned with the natural directions of variation in $\mathcal{R}$. The relationship between κ, α, and the theoretical face manifold capacity is analyzed in Appx. C (Sec. C.8).

Identity Construction and Hard Check. A candidate virtual identity is constructed as:

$$s = \text{normalize}(r + \alpha z), \tag{7}$$

where $r \in \mathcal{R}$ is the reference centroid, z is from Eq. (6), and $\alpha > 0$ controls perturbation strength.

Lemma 1 (Effect of Perturbation Strength)**.** *For any $r, z \in \mathbb{S}^{d-1}$, the cosine similarity between $s = \text{normalize}(r + \alpha z)$ and r satisfies:*

$$\cos(s,\, r) = \frac{1 + \alpha\,(r \cdot z)}{\sqrt{1 + 2\alpha\,(r \cdot z) + \alpha^2}}. \tag{8}$$

In the special case κ=0, $z = z^*$; as r is positively correlated with its nearest neighbors $\mathcal{N}(r)$ on $\mathbb{S}^{d-1}$, the weighted centroid $\sum_k w_{n_k} c_{n_k}$ is positively aligned with r, giving $r \cdot z^* < 0$, and $\cos(s, r)$ is strictly decreasing in α. In the orthogonal case $r \perp z$, Eq. (8) simplifies to $\cos(s, r) = 1/\sqrt{1+\alpha^2}$.

The parameter α controls the fundamental trade-off in BIP, as quantified in Tab. 1: larger α increases angular displacement from real identity territories improving non-collision, while smaller α keeps provisioned identities closer to the face manifold improving realizability. Each candidate s is accepted only when it satisfies the BIP constraints of Definition 1 with respect to the current state of the gallery:

$$\cos(s,\, c_i) < \tau \quad \forall\, i \in \{1, \ldots, M\} \quad \text{(non-collision with } \mathcal{R}\text{)}, \tag{9}$$

$$\cos(s,\, v_j) < \tau \quad \forall\, v_j \in \mathcal{V}_t \quad \text{(separability within } \mathcal{V}_t\text{)}, \tag{10}$$

where $\mathcal{V}_t$ denotes the set of accepted virtual identities at step t; failed candidates are resampled with fresh η. By induction, if every candidate passes these checks before acceptance, the final $\mathcal{V}$ satisfies Eqs. (2)–(3) by construction. Appx. C (Corollary 1) establishes that $\alpha > \sqrt{1-\tau^2}/\tau$ is sufficient for non-collision in the repulsive case; the full α ablation is in Tab. 1. The theoretical capacity of the face manifold vastly exceeds any foreseeable provisioning count; we demonstrate 10M non-colliding virtual identity embeddings against $|\mathcal{R}|$=360K with no observed collision at $\tau = 0.391$. For open-world robustness against subsequently enrolled real identities, a safety buffer $\tau_{\text{safe}} = \tau - \Delta$ can be applied at provisioning time; Appx. C (Proposition 2) formalizes this guarantee. Fig. 3 shows the three components: repulsion direction, PCA-aware perturbation, and hard non-collision check.

3.3 Identity Realization via GapGen

A provisioned embedding $s \in \mathcal{V}$ defines a target location in the biometric space; **GapGen** realizes it as a face image $\tilde{x}$ such that $\phi(\tilde{x})$ remains faithful to s. The challenge is that, since s is intentionally displaced from the real identity distribution, a pretrained generator G, optimized on real embeddings, exhibits reduced fidelity when conditioned on such out-of-distribution inputs.

Identity-conditioned Generator. We build on InstantID [34] as the identity-conditioned generator G, which accepts an ArcFace embedding as conditions and generates 1024×1024 face images, denoted as $\tilde{x} = G(s)$. Since BIP uses the same ArcFace embedding space for both allocation and generation, provisioned embeddings $s \in \mathcal{V}$ can be passed directly as conditions without additional projection. Full implementation details are in Appx. D.

GapGen Fine-tuning. We fine-tune G with a curriculum that interleaves *real steps* and *virtual steps* at mixing ratio $\beta \in [0, 1]$. Both step types share the round-trip identity loss:

$$\mathcal{L}_{\text{RT}}(\tilde{x},\, e^*) = 1 - \cos(\phi(\tilde{x}),\, e^*)\,, \tag{11}$$

where e^* is the target identity embedding and $\rho_{\text{RT}} = \cos(\phi(\tilde{x}), e^*)$ is the *round-trip similarity* quantifying rendering fidelity.

Real steps train G on real images x from FFHQ [14], chosen for its 1024×1024 resolution consistent with our generation target, with target $e^* = \phi(x)$:

$$\mathcal{L}_{\text{real}} = \mathcal{L}_{\text{denoise}} + \lambda_{\text{id}}\, \mathcal{L}_{\text{RT}}(\tilde{x},\, \phi(x)) + \lambda_{\text{perc}}\, \mathcal{L}_{\text{perc}}, \tag{12}$$

where $\mathcal{L}_{\text{denoise}} = \mathbb{E}_{t,\epsilon}[\|\epsilon - \epsilon_\theta(x_t, t, \phi(x))\|^2]$ is the standard noise prediction loss [11], and $\mathcal{L}_{\text{perc}}$ is a perceptual similarity loss [41]. Real steps adapt G to the high-resolution portrait domain and preserve identity fidelity on in-distribution embeddings.

Virtual steps address the core challenge of BIP realization: provisioned embeddings $s \in \mathcal{V}$ have no paired ground-truth images, as virtual identities have no physical counterpart. At each virtual step, we sample a reference centroid $r \in \mathcal{R}$, draw $\alpha \sim \text{Uniform}(\alpha_{\min}{=}2, \alpha_{\max}{=}5)$, construct $s = \text{normalize}(r + \alpha z)$ on-the-fly, generate $\tilde{x} = G(s)$ via truncated DDIM sampling, and optimize $\mathcal{L}_{\text{RT}}(\tilde{x}, s)$ (Eq. (11) with $e^*{=}s$). This unsupervised objective drives G to extend its generative capability into the non-collision region of the embedding space, *without using paired training data*.

Intra-class Variation. Multiple images per virtual identity are generated by varying facial pose, lighting conditions, photographic style, and background through different pose configurations and text style prompts, producing within-identity variation comparable to unconstrained portrait photography.

3.4 Unified Recognition and Real-vs-Virtual Detection

The BIP criteria of Sec. 3.1 evaluate virtual identities: whether they are non-colliding, separable, and realizable. Deploying digital entities alongside real humans raises a complementary set of questions that no existing benchmark addresses: can virtual identities be verified against themselves? How do they interact with real identities under standard recognition protocols? Can a deployed system reliably distinguish real from virtual? We construct **v-LFW** to open this evaluation frontier.

Benchmark Construction. We construct **v-LFW** to mirror LFW [12] in both identity count (5,749) and image count (13,233), supporting controlled evaluation under established face recognition protocols. Each virtual identity is provisioned against $\mathcal{R}$ augmented with all LFW identities, ensuring non-collision with the benchmark. Images are rendered at 1024×1024 with diverse pose and style variation, then filtered for face quality. We define four protocols: virtual face verification, cross-reality matching, real-vs-virtual detection, and unified recognition/detection.

IAPCT: A Diagnostic Evaluation Tool. To support the real-vs-virtual detection and unified recognition protocols of v-LFW, we provide **IAPCT** (Identity-Anchored Patch Consistency Transformer) as a lightweight diagnostic tool built on the frozen encoder ϕ; it is not a primary contribution of this work. Identity-conditioned generators match the target embedding at the output level but impose no constraint on intermediate feature maps, creating detectable inconsistencies between local patch statistics and the global identity embedding $e = \phi(x)$ that IAPCT exploits. Let $e = \phi(x) \in \mathbb{S}^{d-1}$ denote the identity embedding from the frozen backbone. Multi-scale spatial tokens $\{t_{l,p}\}$ are extracted from four intermediate feature maps $\{F_l\}_{l=1}^{4}$ and projected to d_{model}=256 via per-stage linear layers. The projected identity anchor $e_{\text{proj}} = W_{\text{id}}\, e$ serves as the cross-attention key in a 4-layer transformer encoder [32, 8]; each layer applies self-attention among spatial tokens followed by cross-attention to e_{proj}, producing patch-identity consistency scores:

$$\gamma_{l,p} = \sigma\left(\frac{(W_Q\, t_{l,p}) \cdot (W_K\, e_{\text{proj}})}{\sqrt{d_k}}\right) \in (0,1), \tag{13}$$

where σ is the sigmoid function and $\gamma_{l,p}$ measures the consistency of patch p at stage l with the claimed identity e. Real faces yield uniformly high $\gamma_{l,p}$ since every spatial region originates from the same physical person; virtual identity images yield heterogeneous $\gamma_{l,p}$ due to unconstrained intermediate features. A `[CLS]` token produces $\hat{y}$ via an MLP head with loss $\mathcal{L}$=BCE$(\hat{y}, y)$+$\lambda_c(H(\gamma^{\text{virtual}})-H(\gamma^{\text{real}}))$, where $H(\gamma^{(x)})=-\sum_{l,p}\bar{\gamma}_{l,p}^{(x)}\log\bar{\gamma}_{l,p}^{(x)}$ is the entropy of the normalized attention distribution and ϕ remains frozen. The identity embedding e=$\phi(x)$ simultaneously serves face verification at no additional backbone cost. Full details are in Appx. E.

4 Experiments

Baselines. We compare BIP with representative synthetic face generation methods: *DCFace* [16], a dual-condition diffusion model optimized for inter-class separability among synthetics; *Arc2Face* [24], which conditions generation on real ArcFace embeddings; and *Vec2Face* [37], which synthesizes identity-consistent faces from target embeddings with explicit separability control. All baselines are designed for generating face recognition training data, not biometric identity provisioning.

BIP Configuration. We use ArcFace ResNet-100 [7] trained on Glint360K [1] as ϕ (d=512, M=360,232). Unless otherwise stated, we set τ=0.391 (FAR$\approx 2\times10^{-5}$ on IJB-B; Appx. B), α=4.0, K=10 nearest neighbors, temperature t=0.1, and PCA noise weight κ=1.0. GapGen is fine-tuned from InstantID [34] with mixing ratio β=0.2, λ_{id}=0.1, λ_{perc}=0.1, generating faces of 1024×1024 pixels. The large-scale setting provisions ***10 million*** virtual identities; image realization is validated at ***1 million*** identities. Additional details are in the corresponding sections of the Appx.

Geometry and Scale: Non-Collision Holds at Ten Million. Tab. 1 jointly ablates perturbation strength α, provisioning scale $|\mathcal{V}|$, and verification threshold τ across the full IJB-B operating range (see Fig. 6 in Appx. B). At the primary threshold τ=0.391 and $\alpha\geq4$, both Non-Collision and Inter-Sep reach **100%/100% at 100K, 1M, and 10M scale** — ten million non-colliding virtual identities provisioned against 360K real identities with zero observed collision — consistent with the theoretical sufficient bound $\alpha^*(0, 0.391)\approx2.35$ (Corollary 1). α=2 falls below this threshold and yields only 85.6% Non-Collision, while $\alpha\geq3$ already recovers near-perfect performance, confirming the theoretical prediction. Scaling from 100K to 10M incurs negligible cost: Non-Collision remains at 100% throughout, and Inter-Sep stays above 99.98% at τ=0.391, demonstrating that BIP allocation

Table 1: BIP Non-Collision / Inter-Class Sep (%) across identity scale $|\mathcal{V}|$, perturbation strength α, and threshold τ. At the operating point τ=0.391, $\alpha \geq 4$ achieves 100%/100% values at both 100K and 10M scale.

$\lvert\mathcal{V}\rvert$	α	$\tau = 0.448$	$\tau = \mathbf{0.391}$	$\tau = 0.360$	$\tau = 0.341$	$\tau = 0.330$	$\tau = 0.319$
100K	2	100.00/100.00	85.63/100.00	64.57/99.99	48.69/99.93	39.05/99.81	30.55/99.54
	3	100.00/100.00	99.98/100.00	99.80/99.99	98.96/99.93	97.46/99.81	94.28/99.55
	4	100.00/100.00	**100.00 / 100.00**	99.91/99.99	99.39/99.94	98.19/99.82	95.55/99.54
	5	100.00/100.00	100.00/100.00	99.91/99.99	99.40/99.94	98.23/99.81	95.62/99.53
1M	2	100.00/100.00	85.61/100.00	64.48/99.92	48.51/99.42	39.01/98.24	30.42/95.48
	3	100.00/100.00	99.98/100.00	99.78/99.93	98.94/99.43	97.42/98.28	94.20/95.54
	4	100.00/100.00	**100.00 / 100.00**	99.91/99.92	99.38/99.43	98.17/98.27	95.52/95.53
	5	100.00/100.00	100.00/100.00	99.91/99.92	99.38/99.43	98.22/98.27	95.53/95.54
10M	2	100.00/100.00	85.10/99.97	64.74/99.22	48.06/94.51	39.00/85.55	30.16/66.62
	3	100.00/100.00	99.98/99.98	99.77/99.21	98.99/94.67	97.39/85.50	94.16/67.06
	4	100.00/100.00	**100.00 / 99.98**	99.91/99.24	99.39/94.66	98.27/85.57	95.36/66.99
	5	100.00/100.00	100.00/99.98	99.91/99.23	99.40/94.64	98.28/85.53	95.38/66.94

Table 2: Synthetic face generation under BIP criteria. All methods are re-encoded by the same frozen ArcFace, tested against $\mathcal{R}$ (M=360K) at τ=0.391. FID measures perceptual image quality against FFHQ. RT pass rate reports the fraction of generated images whose re-encoded embedding satisfies ρ_{RT}>0.6, quantifying how faithfully the generator realizes the target identity embedding.

Method	#Identity	Non-Collision ↑	Inter-Sep ↑	FID ↓	RT pass rate ↑	Resolution
CAS. (Real) [40]	10.6K	0.32	90.08	131.70	-	250×250
DCFace [16]	10K	72.38	49.20	139.70	-	112×112
Arc2Face [24]	10K	93.90	88.67	74.11	0.00	512×512
Vec2Face [37]	10K	53.92	65.02	159.30	62.78	112×112
Vec2Face [37]	100K	52.95	45.44	159.18	61.43	112×112
Vec2Face [37]	500K	52.93	36.22	159.29	62.01	112×112
BIP+GapGen (ours)	10K	**98.43**	**99.57**	56.20	87.94	1024×1024
BIP+GapGen (ours)	100K	98.38	97.84	**54.63**	**89.42**	1024×1024
BIP+GapGen (ours)	500K	98.24	86.30	55.62	88.31	1024×1024
BIP+GapGen (ours)	1M	98.07	78.36	56.47	88.75	1024×1024

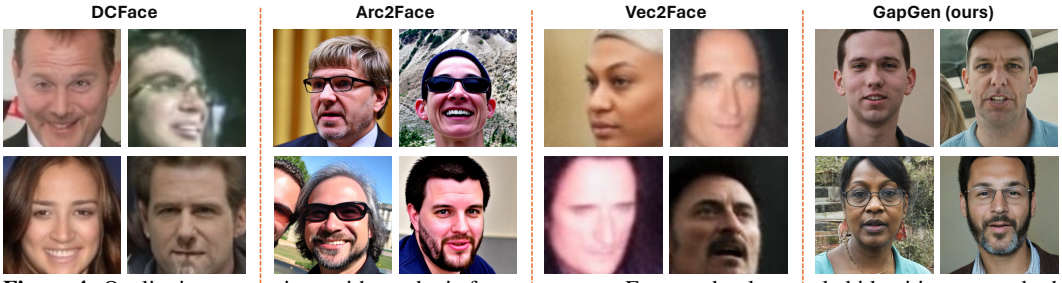


Figure 4: Qualitative comparison with synthetic face generators. Four randomly sampled identities per method. GapGen produces photorealistic 1024×1024 portrait faces with natural texture and coherent lighting, free from blurring, distortion, and uncontrolled backgrounds of the baselines.

is not capacity-limited at any scale tested. At stricter thresholds (smaller τ), Inter-Sep degrades more noticeably at 10M (*e.g.*, 66.99% at τ=0.319), consistent with the tighter capacity headroom at extreme operating points (Appx. C); τ=0.391 remains robust across all scales.

Realization Quality: GapGen Outperforms Baselines across All BIP Criteria. Tab. 2 compares BIP+GapGen against baselines under the three BIP criteria at τ=0.391. While BIP allocation scales to 10M embeddings, image realization is validated up to 1M virtual images due to the computational cost of 1024×1024 generation; this reflects a deliberate fidelity-scale trade-off rather than an allocation bottleneck. Existing methods fail on at least one criterion: DCFace [16] reaches only 49.20% Inter-Sep; Vec2Face [37] (reproduced from released code) achieves only 53.92% Non-Collision at 10K with Inter-Sep collapsing to 36.22% at 500K; Arc2Face [24] achieves 93.90% Non-Collision but at ρ_{RT}=0.00, indicating complete failure of embedding fidelity after re-encoding. BIP+GapGen achieves 98.43% Non-Collision, 99.57% Inter-Sep, FID 56.20, and round-trip pass rate 87.94% at 10K, the best across all criteria at 9× higher resolution than baselines. As scale increases, Non-Collision remains stable (98.07% at 1M); Inter-Sep decreases to 78.36%, reflecting the realization gap between embedding-space allocation (Tab. 1, Inter-Sep ≈ 100% by construction) and image-level realization at scale. Fig. 4 confirms the qualitative advantage; additional visual results are in Appx. D.

Deepfake Detectability of Virtual Faces. A concern with high-fidelity virtual face generation is whether provisioned identities could be misused as deepfakes or synthetic personas. We evaluate this

Table 3: (a) Five SoTA deepfake detectors [20] applied zero-shot to BIP-generated virtual faces. **(b)** v-LFW evaluation on the combined LFW+v-LFW set. R-V pairs are exclusively impostor pairs by BIP construction. Unified denotes joint recognition and detection on the combined set. "–": task not supported by the method.

Detector	ACC ↑
SPSL [21]	87.76
UCF [39]	93.14
Xception [27]	95.13
EfficientB4 [31]	97.13
PG-FDD [19]	**99.57**

(a)

Metric	DCFace [16]+AdaFace	Vec2Face [37]+ArcFace	BIP+IAPCT (Ours)
R-R Acc ↑	98.58	98.87	**99.80**
V-V Acc ↑	98.85	99.04	**99.18**
R-V FAR ↓	0.05	0.02	**0.01**
Det. AUC ↑	–	–	**98.13**
Unified ↑	–	–	**99.20**

(b)

by applying five SoTA deepfake detectors to our generated virtual faces zero-shot, without fine-tuning on virtual faces; all detectors are trained on AIFaceFairnessBench [20]. As in Tab. 3a, all detectors achieve high accuracy on BIP-generated faces, confirming that existing safety infrastructure already covers provisioned virtual identities without modification. The non-collision guarantee and deepfake detectability are complementary: the former ensures virtual identities are biometrically distinct from all real humans, while the latter confirms they remain within the reach of existing forensic systems.

Open-World Non-Collision as Real Galleries Grow. The hard checks guarantee non-collision against $\mathcal{R}$ by construction, but provide no guarantee against real identities enrolled after provisioning. To stress-test this, we use 180K held-out identities from WebFace4M [42] (disjoint from $\mathcal{R}$) and fix $|\mathcal{V}|$ at one million virtual identities, measuring the per-pair collision rate $C/(N\times L)$ as $|\mathcal{R}_{\text{test}}|/|\mathcal{R}|$ grows from 0.1 to 0.5 ($|\mathcal{R}|$=360K). Fig. 5 shows flat curves across all α values at τ=0.391, confirming p_{coll} is a stable geometric property independent of test-gallery scale (Appx. C). At our default α=4.0, collision rate is stable near zero, demonstrating genuine open-world non-collision beyond $\mathcal{R}$.

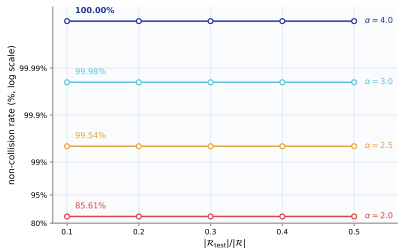


Figure 5: Open-world non-collision. At $\alpha\geq$3.0, collision is near zero in all scales.

Cross-Reality Recognition on v-LFW Dataset. A deployed system serving both real humans and digital entities must simultaneously answer two questions for any face presented to it: *who is this?* and *is this a real or virtual identity?* v-LFW is the first benchmark to answer both questions jointly, mirroring LFW in identity count (5,749) and image count (13,233) and enabling five protocols on the combined LFW+v-LFW set. **R-R** and **V-V** follow the standard LFW verification protocol; **R-V** pairs are exclusively impostor pairs by BIP construction; **Detection** distinguishes real from virtual without identity labels; **Unified** simultaneously produces a verified identity and a real-vs-virtual decision. As shown in Tab. 3b, BIP + IAPCT is the only method supporting all five protocols. R-R Acc 99.80% confirms that IAPCT leaves recognition performance fully intact. R-V FAR $\approx$ 0% validates the non-collision guarantee end-to-end at image level, and Detection AUC 98.13% confirms real and virtual populations are reliably separable without modifying the frozen backbone.

5 Conclusion

We introduced Biometric Identity Provisioning (BIP), the first formal framework for allocating non-colliding biometric identities to digital entities at scale. The key geometric insight is that valid virtual identities must occupy unclaimed gaps within the real face manifold rather than a separate free subspace. This finding shapes repulsion-based allocation, gap-aware realization via GapGen, and the capacity analysis establishing that the face manifold supports vastly more non-colliding positions than any foreseeable enrollment scale. Against a gallery of 360K real identities, 10M non-colliding virtual identity embeddings are provisioned with no observed collision and realized as high-fidelity face images with end-to-end non-collision verified by re-encoding. Our v-LFW dataset benchmarks real-virtual coexistence; R-V FAR $\approx$ 0% confirms the BIP guarantee at image level, and zero-shot deepfake detectability confirms no new forensic risks are introduced.

Limitations. BIP guarantees non-collision against $\mathcal{R}$ at provisioning time; protection against subsequently enrolled identities requires the safety buffer (Appx. C) and periodic revocation. Capacity bounds rest on a spherical submanifold approximation rather than a precise characterisation of the face manifold (Appx. C). GapGen fidelity degrades at large α, and all experiments use ArcFace ResNet-100; generalization to other encoders remains to be evaluated.

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Appendix

This appendix provides full details supporting the main paper.

- Sec. A: Broader Impact and Ethics.
- Sec. B: Verification threshold selection and IJB-B operating points.
- Sec. C: Capacity bound derivations.
  - Sec. C.1: PCA analysis and submanifold model.
  - Sec. C.2: Spherical geometry preliminaries and GV lower bound.
  - Sec. C.3: Gilbert–Varshamov lower bound.
  - Sec. C.4: Effective capacity, minimum separation threshold (Corollary 1), and sensitivity to τ.
  - Sec. C.5: Repulsion direction analysis.
  - Sec. C.6: Safety buffer (Proposition 2).
  - Sec. C.7: Empirical capacity estimator.
  - Sec. C.8: Capacity, α, and PCA noise.
- Sec. D: Additional implementation Details of G (GapGen).
  - Base pipeline, conditioning, and sampling.
  - Round-trip re-encoding.
  - Gap-aware fine-tuning.
- Sec. E: Verification, Visual Examples, and IAPCT.
  - Sec. E.1: v-LFW protocol and visual examples.
  - Sec. E.2: IAPCT architecture, tokenisation, and training details.
- Sec. F: Additional results.
  - Sec. F.1: t-SNE of $\mathcal{R}$ vs. $\mathcal{V}$.
  - Sec. F.2: Additional image grids.

A Broader Impact and Ethics

BIP provisions biometric identities for digital entities, not synthetic faces of real people. The non-collision constraint ensures provisioned identities lie outside all enrolled real identity territories, and Sec. 4 confirms they are detectable by existing deepfake detectors without modification. Potential misuse to evade biometric access control is mitigated by the gallery-relative design: an adversary requires access to the full enrollment gallery. All training data are used under their respective licenses, and v-LFW contains no real person's likeness.

B Verification Threshold Selection

The BIP non-collision constraint $\cos(v_j, c_i) < \tau$ and all capacity bounds in Appendix C are stated in terms of a verification threshold τ that must be grounded in the operating point of the deployed recognition system rather than set arbitrarily.

Protocol. We evaluate ArcFace ResNet-100 [7] (Glint360K, `antelopev2/glintr100`, ONNX) on IJB-B [35] under the 1:1 verification protocol across a sweep of FAR operating points from 10^{-5} to 10^{-4}. The ROC curve and operating points are shown in Fig. 6.

Threshold Choice and BIP Implications. The choice of τ directly governs the stringency of the BIP non-collision constraint and the available face manifold capacity:

- **Larger τ (stricter recognition, *e.g.*, FAR $= 10^{-5}$, $\tau = 0.448$):** fewer impostor pairs are falsely accepted. The acceptance cap $\{x : \cos(x, c_i) \geq \tau\}$ around each real identity is *smaller*, and the BIP non-collision constraint $\cos(v_j, c_i) < \tau$ is looser.
- **Smaller τ (more permissive recognition, *e.g.*, FAR $= 10^{-4}$, $\tau = 0.319$):** more impostor pairs are accepted. The acceptance cap around each real identity is *larger*, and the BIP non-collision constraint is stricter, requiring more perturbation.

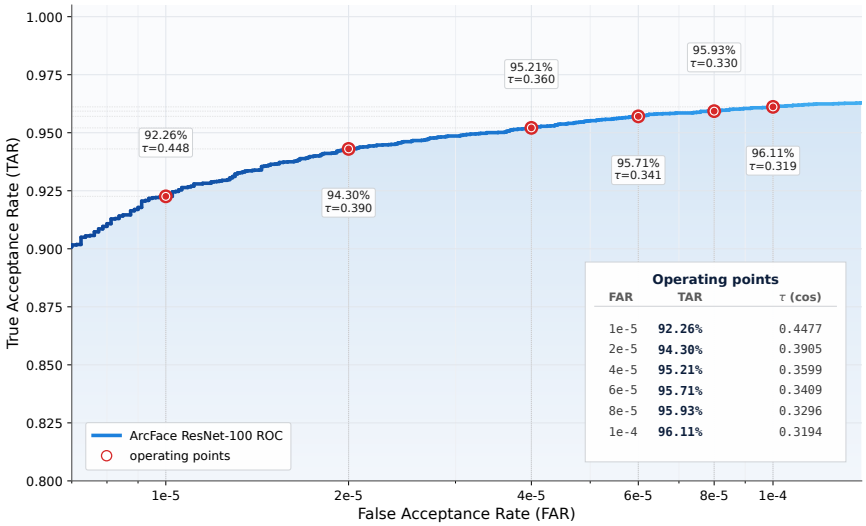


Figure 6: ROC curve for ArcFace ResNet-100 on IJB-B 1:1 verification (8.01M pairs; 10,270 genuine / 8.00M impostor; AUC $= 99.49\%$). Red circles mark the six operating points used as BIP threshold candidates; τ denotes the cosine similarity threshold achieving the stated FAR.

We use $\tau = 0.391$ as the primary operating point in all main experiments, corresponding to FAR $\approx 2 \times 10^{-5}$ and TAR $= 94.30\%$ on IJB-B. At this operating point, one in 5×10^4 impostor pairs is falsely accepted, a conservative threshold appropriate for high-security deployments. We ablate BIP metrics across all six operating points in Tab. 1 of the main paper, demonstrating that our method is robust to threshold choice within FAR $\in [10^{-5}, 10^{-4}]$.

C Capacity Bound Derivation

This section provides full derivations supporting the BIP capacity analysis in Sec. 3.2. Throughout, $\mathbb{S}^{d-1}$ denotes the unit hypersphere in $\mathbb{R}^d$ with $d = 512$, $\tau \in (0, 1)$ is the verification threshold of Assumption 1, and $\tilde{d} = 269$ is the effective manifold dimensionality established in Sec. C.1 below. All reported numerical bounds use the *exact* regularized incomplete beta function; the Gaussian Q-function is used for intuition only.

Notation. We distinguish four related quantities used throughout, on distinct mathematical objects:

- $A_{\text{pack}}(\mathcal{M}_{\text{face}}, \tau)$: the true (unknown) maximum packing number of a τ-code on the face manifold.
- $A_{\text{pack}}(\mathbb{S}^{\tilde{d}-1}, \tau)$: the packing number on the ambient sphere of the submanifold approximation, satisfying $A_{\text{pack}}(\mathcal{M}_{\text{face}}, \tau) \leq A_{\text{pack}}(\mathbb{S}^{\tilde{d}-1}, \tau)$ since $\mathcal{M}_{\text{face}} \subset \mathbb{S}^{\tilde{d}-1}$.
- $A_{\text{GV}}(\tilde{d}, \tau) := 1/\mu(\tau, \tilde{d})$: the computable Gilbert–Varshamov lower bound on the ambient-sphere packing number, $A_{\text{pack}}(\mathbb{S}^{\tilde{d}-1}, \tau) \geq A_{\text{GV}}$.
- $A_{\text{eff}}(\tau) := 1/p_{\text{coll}}(\tau)$: the empirical effective capacity, estimated from observed collision counts between $\mathcal{V}$ and unseen real identities under the empirical real-face distribution.

A_{GV} and A_{eff} measure different objects and should not be read as competing bounds on the same quantity; see Sec. C.7 for their distinct roles.

C.1 Real Face Manifold: PCA and Submanifold Model

Fig. 7 visualizes the PCA structure of Glint360K identity centroids, confirming Observation 1: 95% of variance concentrates within $\tilde{d}$=269 components out of d=512. We therefore model $\mathcal{M}_{\text{face}}$ as locally approximated by $\mathbb{S}^{\tilde{d}-1}$ and apply all packing bounds within this reduced sphere. This is a *capacity-scale approximation*: the cap angle $\arccos(0.391) \approx 67°$ is not a small local neighborhood, so results should be interpreted as headroom estimates rather than precise geometric statements.

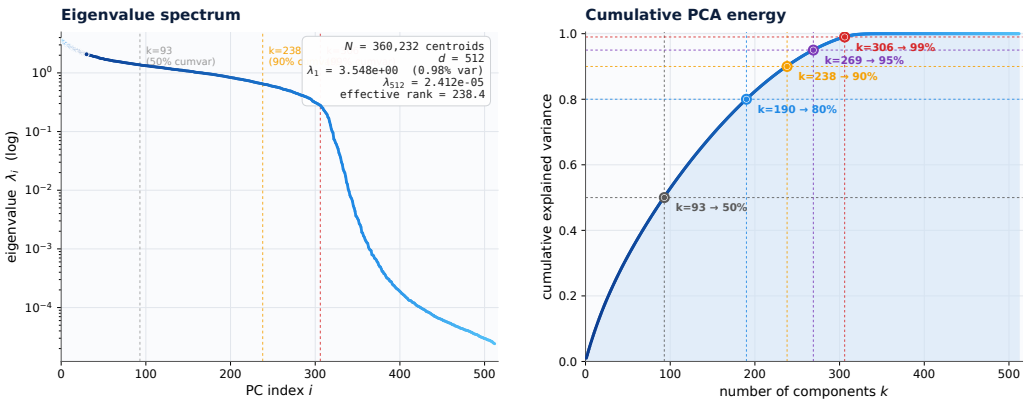


Figure 7: PCA of Glint360K identity centroids (N=360,232, ArcFace ResNet-100, d=512). **Left (eigenvalue spectrum):** λ_k decays sharply after $k\approx$300, with $\lambda_{512} = 2.4\times10^{-5}$ five orders of magnitude below $\lambda_1 = 3.55$; effective rank = 238.4. **Right (cumulative explained variance):** 50% of variance is captured within k=93 components, 90% within k=238, 95% within k=269, and 99% within k=306.

C.2 Spherical Geometry Preliminaries

Spherical Cap Volume. The *spherical cap* of cosine radius τ centered at $p \in \mathbb{S}^{d-1}$ is:

$$\mathrm{Cap}(p,\tau) = \left\{x \in \mathbb{S}^{d-1} : \cos(x,p) \geq \tau\right\}. \tag{14}$$

Its normalized volume is independent of p by rotational symmetry:

$$\mu(\tau,d) = \frac{\mathrm{Vol}(\mathrm{Cap}(p,\tau))}{\mathrm{Vol}(\mathbb{S}^{d-1})} = \frac{1}{2} I_{1-\tau^2}\left(\frac{d-1}{2}, \frac{1}{2}\right), \tag{15}$$

where $I_x(a,b)$ is the regularized incomplete beta function.

Verification of Eq. (15). For $\mathbb{S}^1$ (circle in $\mathbb{R}^2$) with $\tau = 0.5$: $\mu = \frac{1}{2} I_{0.75}(\frac{1}{2},\frac{1}{2}) = \frac{1}{\pi}\arcsin(\sqrt{0.75}) = \frac{1}{3}$, matching the $120°/360°$ arc fraction for a $60°$ half-angle cap. ✓

Monotonicity. $\mu(\tau,d)$ is *strictly decreasing* in τ: larger τ defines a smaller cap. $\mu(\tau,d)$ is also strictly decreasing in d by concentration of measure: for $\tilde{d} < d$, $\mu(\tau,\tilde{d}) \geq \mu(\tau,d)$.

Gaussian Approximation (Intuition Only). $\mu(\tau,d) \approx Q(\tau\sqrt{d})$ by the CLT, but in the far-tail regime this substantially overestimates the exact value. At $\tau = 0.391$, $d = 269$: the Gaussian gives $Q(0.391\sqrt{269}) \approx Q(6.41) \approx 7.2\times10^{-11}$ versus the exact $\mu = 1.35\times10^{-11}$ (a 5.3× overestimate). All reported bounds use Eq. (15) directly.

Packing Number. A *spherical τ-code* is a set $\mathcal{S} \subset \mathbb{S}^{d-1}$ with $\cos(s_i,s_j) < \tau$ for all $i \neq j$. Its maximum cardinality is the packing number:

$$A(d,\tau) = \max\left\{|\mathcal{S}| : \mathcal{S} \subset \mathbb{S}^{d-1},\ \cos(s_i,s_j) < \tau\ \forall i \neq j\right\}, \tag{16}$$

which equals the BIP capacity of Definition 1 when $\mathcal{R} = \emptyset$.

C.3 Gilbert–Varshamov Lower Bound

Theorem 1 (Gilbert–Varshamov Bound)**.**

$$A(d,\tau) \ \geq\ \frac{1}{\mu(\tau,d)} = A_{\mathrm{GV}}(d,\tau). \tag{17}$$

Proof. Let $\mathcal{S}^* \subset \mathbb{S}^{d-1}$ be a maximal τ-code, *i.e.*, no additional point can be appended without violating the separation constraint. Maximality implies the caps $\{\mathrm{Cap}(s,\tau)\}_{s\in\mathcal{S}^*}$ cover all of $\mathbb{S}^{d-1}$; otherwise an uncovered point could be appended, contradicting maximality. Comparing total cap volume to sphere volume:

$$|\mathcal{S}^*| \cdot \mu(\tau,d) \geq 1 \ \implies\ |\mathcal{S}^*| \geq \frac{1}{\mu(\tau,d)}. \tag{18}$$

Since $\mathcal{S}^*$ is itself a valid τ-code and $A(d,\tau)$ is the maximum cardinality over all such codes: $A(d,\tau) \geq |\mathcal{S}^*| \geq \frac{1}{\mu(\tau,d)}$. □

Numerical Evaluation at $\tau = 0.391$, $d = 512$.

$$\mu(0.391, 512) \approx 1.74 \times 10^{-20}, \quad A_{\mathrm{GV}}(512, 0.391) \approx 5.75 \times 10^{19} \approx 2^{65.6}. \tag{19}$$

This full-hypersphere bound is not directly useful for BIP since real face identities occupy only a $\tilde{d}$-dimensional submanifold of $\mathbb{S}^{511}$.

C.4 Effective Capacity and GV Lower Bound

We use Theorem 1 only to obtain an ambient-sphere capacity reference for the $\mathbb{S}^{\tilde{d}-1}$ approximation. This reference should not be interpreted as a rigorous lower bound on the packing capacity of $\mathcal{M}_{\text{face}}$.

Proposition 1 (Ambient-sphere packing under the submanifold model)**.** *Under the submanifold approximation $\mathcal{M}_{face} \subset \mathbb{S}^{\tilde{d}-1}$ of Sec. C.1, the packing number of the ambient sphere satisfies:*

$$A_{\mathrm{pack}}(\mathbb{S}^{\tilde{d}-1}, \tau) \;\geq\; A_{\mathrm{GV}}(\tilde{d}, \tau) = \frac{1}{\mu(\tau, \tilde{d})} = \frac{2}{I_{1-\tau^2}\left(\frac{\tilde{d}-1}{2}, \frac{1}{2}\right)}. \tag{20}$$

For $\tilde{d} = 269$ and $\tau = 0.391$, $\mu(0.391, 269) = 1.35 \times 10^{-11}$, giving:

$$A_{\mathrm{GV}}(269, 0.391) = 7.41 \times 10^{10} \approx 2^{36.1}. \tag{21}$$

Since $\mathcal{M}_{face} \subset \mathbb{S}^{\tilde{d}-1}$, we have $A_{\mathrm{pack}}(\mathcal{M}_{face}, \tau) \leq A_{\mathrm{pack}}(\mathbb{S}^{\tilde{d}-1}, \tau)$. We therefore use A_{GV} only as an ambient-sphere capacity-scale reference, not as a rigorous lower bound on $A_{\mathrm{pack}}(\mathcal{M}_{face}, \tau)$.

Proof. Apply Theorem 1 to dimension $\tilde{d}$ in place of d; the inequality on $A_{\mathrm{pack}}(\mathcal{M}_{\text{face}}, \tau)$ follows from the subset relation $\mathcal{M}_{\text{face}} \subset \mathbb{S}^{\tilde{d}-1}$. □

Corollary 1 (Minimum Separation Threshold)**.** *For $r, z \in \mathbb{S}^{d-1}$ with $p = r \cdot z$ and $|p| < \tau$, let $s = \mathrm{normalize}(r + \alpha z)$. The unique $\alpha^* > 0$ at which $\cos(s, r) = \tau$ exactly is:*

$$\alpha^*(p, \tau) = \frac{\sqrt{1-\tau^2}\left[p\sqrt{1-\tau^2} + \tau\sqrt{1-p^2}\right]}{\tau^2 - p^2}. \tag{22}$$

For $\alpha > 0$ and $|p| < 1$, $\cos(s, r)$ is strictly decreasing in α:

$$\frac{d}{d\alpha}\frac{1+\alpha p}{\sqrt{1+2\alpha p+\alpha^2}} = \frac{\alpha(p^2-1)}{(1+2\alpha p+\alpha^2)^{3/2}} < 0, \tag{23}$$

so $\cos(s, r) < \tau$ requires $\alpha > \alpha^*(p, \tau)$ strictly.

In the orthogonal case $p = 0$, Eq. (22) simplifies to:

$$\alpha^*(0, \tau) = \frac{\sqrt{1-\tau^2}}{\tau}. \tag{24}$$

For the operating thresholds used in this work ($\tau \leq 0.5 < 1/\sqrt{2}$), $\alpha^*(0, \tau)$ is the maximum of $\alpha^*(p, \tau)$ over the repulsive range $p \in [-\tau, 0]$. Therefore, $\alpha > \sqrt{1-\tau^2}/\tau$ is sufficient for $\cos(s, r) < \tau$ over the repulsive range $p \in [-\tau, 0]$.

Scope. *Corollary 1 bounds $\cos(s, r) < \tau$ only against the construction reference r. The orthogonal worst-case $\alpha^*(0, \tau)$ should therefore be read as a principled lower bound on α in the proposal distribution. Gallery-wide non-collision in embedding space is not guaranteed by α alone; it is enforced exactly by the hard check of Eq. (9) at acceptance time. Image-level non-collision after realization, and non-collision against unseen future identities, are empirical evaluations reported in Sec. 4 and Sec. C.7.*

Note on positive p: *With PCA-aware noise (Eq. (6)), $p = r \cdot z$ can be positive. For $p > 0$, α^* grows rapidly: $\alpha^*(0.3, 0.391) \approx 9.5$, far exceeding the orthogonal bound 2.35. The hard checks of Eqs. (9)–(10) enforce BIP constraints irrespective of the sign of $r \cdot z$.*

Proof. **Deriving $\alpha^*(p,\tau)$.** From Lemma 1:

$$\cos(s,r) = \frac{1+\alpha p}{\sqrt{1+2\alpha p+\alpha^2}}. \tag{25}$$

Setting $\cos(s,r) = \tau > 0$: the right-hand side equals $\tau > 0$, so the numerator $1+\alpha p$ must be positive at any solution (since the denominator is always positive). Squaring the equality at the solution is therefore valid:

$$\tau^2(1+2\alpha p+\alpha^2) = (1+\alpha p)^2, \tag{26}$$

which rearranges to the quadratic:

$$\alpha^2(\tau^2-p^2) + 2\alpha p(\tau^2-1) + (\tau^2-1) = 0. \tag{27}$$

Since $|p| < \tau$, we have $\tau^2 - p^2 > 0$, so Eq. (27) is a well-posed quadratic. Its discriminant is:

$$\begin{aligned}\Delta &= 4p^2(\tau^2-1)^2 - 4(\tau^2-p^2)(\tau^2-1) = 4(\tau^2-1)\left[p^2(\tau^2-1) - (\tau^2-p^2)\right] \\ &= 4(\tau^2-1)\tau^2(p^2-1) = 4\tau^2(1-\tau^2)(1-p^2) > 0,\end{aligned} \tag{28}$$

giving $\sqrt{\Delta} = 2\tau\sqrt{(1-\tau^2)(1-p^2)}$. The positive root of (27) is:

$$\alpha^*(p,\tau) = \frac{p(1-\tau^2) + \tau\sqrt{(1-\tau^2)(1-p^2)}}{\tau^2-p^2} = \frac{\sqrt{1-\tau^2}\left[p\sqrt{1-\tau^2} + \tau\sqrt{1-p^2}\right]}{\tau^2-p^2}, \tag{29}$$

establishing Eq. (22). At $p=0$: $\alpha^*(0,\tau) = \tau\sqrt{1-\tau^2}/\tau^2 = \sqrt{1-\tau^2}/\tau$, confirming Eq. (24).

Strict inequality Eq. (23). Direct differentiation of $f(\alpha) = (1+\alpha p)/\sqrt{1+2\alpha p+\alpha^2}$:

$$f'(\alpha) = \frac{p\sqrt{1+2\alpha p+\alpha^2} - (1+\alpha p)\cdot\dfrac{p+\alpha}{\sqrt{1+2\alpha p+\alpha^2}}}{1+2\alpha p+\alpha^2} = \frac{\alpha(p^2-1)}{(1+2\alpha p+\alpha^2)^{3/2}}. \tag{30}$$

Since $\alpha > 0$ and $p^2 < 1$ (as $|p| < \tau < 1$), $f'(\alpha) < 0$, establishing Eq. (23). Since $f(0) = 1 > \tau$ and $\lim_{\alpha\to\infty} f(\alpha) = p < \tau$ (as $|p| < \tau$), strict monotonicity of f implies the solution $\alpha^*(p,\tau)$ is unique.

Upper bound at $\tau \le 1/\sqrt{2}$. By the implicit function theorem applied to $f(\alpha^*(p), p) = \tau$, with $f_\alpha < 0$ (from Eq. (23)) and $f_p = \alpha^2(\alpha+p)/(1+2\alpha p+\alpha^2)^{3/2}$:

$$\frac{d\alpha^*}{dp} = -\frac{f_p}{f_\alpha} = \frac{\alpha^*(\alpha^*+p)}{1-p^2}. \tag{31}$$

The sign of (31) depends on $\alpha^*(p,\tau)+p$. Taking the limit $p \to -\tau^+$ using Eq. (22):

$$\lim_{p\to-\tau^+}(\alpha^*(p,\tau)+p) = \frac{1}{2\tau} - \tau = \frac{1-2\tau^2}{2\tau} \;\ge\; 0 \quad\Longleftrightarrow\quad \tau \le \frac{1}{\sqrt{2}}. \tag{32}$$

At $p=0$: $\alpha^*(0,\tau)+0 = \sqrt{1-\tau^2}/\tau > 0$. Since $\alpha^* > 0$ and $1-p^2 > 0$, the sign of $d\alpha^*/dp$ is determined by α^*+p. Using Eq. (22):

$$\alpha^*(p,\tau)+p = \frac{\sqrt{1-p^2}\left[p\sqrt{1-p^2} + \tau\sqrt{1-\tau^2}\right]}{\tau^2-p^2}. \tag{33}$$

The denominator $\tau^2-p^2 > 0$ for $|p| < \tau$, and $\sqrt{1-p^2} > 0$, so the sign reduces to $p\sqrt{1-p^2} + \tau\sqrt{1-\tau^2}$. Define $h(p) = p\sqrt{1-p^2}$; then $h'(p) = (1-2p^2)/\sqrt{1-p^2} \ge 0$ for $p^2 \le \tau^2 \le 1/2$ (since $\tau \le 1/\sqrt{2}$), so h is non-decreasing on $[-\tau, 0]$ and $h(p) \ge h(-\tau) = -\tau\sqrt{1-\tau^2}$. Therefore $p\sqrt{1-p^2} + \tau\sqrt{1-\tau^2} \ge 0$, giving $\alpha^*(p,\tau)+p \ge 0$ and $d\alpha^*/dp \ge 0$ throughout $p \in [-\tau, 0]$. Hence $\alpha^*(0,\tau) = \sqrt{1-\tau^2}/\tau$ is the maximum over the repulsive range. □

Sensitivity to τ. Table 4 reports exact GV bounds and the sufficient perturbation threshold for all thresholds used in this work. Since $\mu(\tau, \tilde{d})$ is strictly decreasing in τ, larger τ corresponds to a less restrictive non-collision constraint (smaller excluded cap per identity), yielding a larger A_{GV}. The safety buffer requires provisioning at $\tau_{\text{safe}} < \tau$ (stricter constraint), which *reduces* A_{GV} and *increases* the required perturbation; see Proposition 2.

Table 4: GV lower bound $A_{\text{GV}} = 1/\mu(\tau, 269)$ on the ambient sphere $\mathbb{S}^{268}$ at the six IJB-B operating points (Fig. 6), computed via exact regularized incomplete beta. $\alpha^*(0,\tau) = \sqrt{1-\tau^2}/\tau$ is the orthogonal worst-case sufficient perturbation for $\cos(s,r) < \tau$ against the construction reference r, over repulsive directions $p \in [-\tau, 0]$, valid for $\tau \leq 1/\sqrt{2}$; gallery-wide non-collision is enforced separately by the hard check of Eq. (9). **Bold**: primary operating point ($\tau = 0.391$, FAR$\approx 2 \times 10^{-5}$).

τ	$\mu(\tau, 269)$	$\log_2 A_{\text{GV}}$	$A_{\text{GV}} = 1/\mu$	$\alpha^*(0,\tau)$
0.319	4.21×10^{-8}	24.50	2.38×10^{7}	2.97
0.330	1.40×10^{-8}	26.09	7.15×10^{7}	2.86
0.341	4.45×10^{-9}	27.74	2.25×10^{8}	2.76
0.360	5.52×10^{-10}	30.75	1.81×10^{9}	2.59
0.391	$\mathbf{1.35 \times 10^{-11}}$	**36.11**	$\mathbf{7.41 \times 10^{10}}$	**2.35**
0.448	4.92×10^{-15}	47.53	2.03×10^{14}	2.00

Ambient-sphere capacity reference. $A_{\text{GV}}(\tilde{d}, \tau)$ lower-bounds the packing number on the ambient sphere $\mathbb{S}^{\tilde{d}-1}$. As noted in Proposition 1, the face-manifold packing $A_{\text{pack}}(\mathcal{M}_{\text{face}}, \tau)$ is upper-bounded by, but not lower-bounded by, $A_{\text{pack}}(\mathbb{S}^{\tilde{d}-1}, \tau)$, so A_{GV} serves as a capacity-scale reference rather than a strict lower bound on $\mathcal{M}_{\text{face}}$ capacity. At $\tau = 0.391$, $A_{\text{GV}}(269, 0.391) \approx 7.41 \times 10^{10}$; both 1M and 10M provisioned identities fall orders of magnitude below this ambient reference:

$$\frac{A_{\text{GV}}(269, 0.391)}{10^6} \approx 2^{16.2}, \qquad \frac{A_{\text{GV}}(269, 0.391)}{10^7} \approx 2^{12.9}. \tag{34}$$

The operationally meaningful capacity statement is empirical: at $\tau = 0.391$ and $\alpha = 4$, no collisions are observed against $\mathcal{R}$ for $|\mathcal{V}|$ up to 10^7 (Tab. 1), and the held-out collision rate against unseen real identities remains stable at ≈ 0 (Fig. 5, Sec. C.7).

C.5 Repulsion Direction: Heuristic Motivation

Remark 2 in the main text introduces z^* as a repulsion heuristic. We provide the precise mathematical analysis here.

What is proven. With $m = \sum_{k=1}^{K} w_{n_k} c_{n_k}$ and $z^* = -m/\|m\|_2$:

$$z^* \cdot c_{n_k} < 0 \quad \text{for any } c_{n_k} \text{ satisfying } c_{n_k}^\top m > 0. \tag{35}$$

Neighbors positively aligned with the weighted centroid m satisfy this condition; whether *all* neighbors do depends on the local geometry and is not guaranteed solely by the nearest-neighbour construction.

What is not proven after normalization. The relevant quantity for BIP is the rate of change of cosine similarity to a neighbor under the spherically-normalized perturbation $s(\alpha) = \text{normalize}(r + \alpha z^*)$. By the chain rule:

$$\frac{d}{d\alpha} \cos(s(\alpha), c_{n_k}) \bigg|_{\alpha=0} = z^* \cdot c_{n_k} - (r \cdot z^*)(r \cdot c_{n_k}). \tag{36}$$

Even when $z^* \cdot c_{n_k} < 0$, the second term $(r \cdot z^*)(r \cdot c_{n_k})$ can dominate: if $r \cdot z^* < 0$ (typical for a repulsion direction) and $r \cdot c_{n_k} > 0$ (neighbor close to r), their product is negative and contributes positively to the derivative. Whether Eq. (36) is negative, *i.e.*, whether moving along z^* genuinely increases cosine distance to c_{n_k} after renormalization — depends on the local geometry and is not guaranteed by the construction of z^* alone.

Correct status. z^* is a *repulsive heuristic*: it improves the probability that a candidate passes the BIP hard checks, but does not formally guarantee it. The formal non-collision guarantee comes exclusively from Eqs. (9)–(10), which accept a candidate s only after exact cosine verification against all of $\mathcal{R}$ and $\mathcal{V}_t$.

C.6 Safety Buffer for Open-World Robustness

Proposition 2 (Safety Buffer). *Let τ be the operating verification threshold (collision when $\cos \geq \tau$) and let $\tau_{\text{safe}} = \tau - \Delta$ with $\Delta > 0$. If $\mathcal{V}$ is provisioned such that $\cos(v_j, c_i) < \tau_{\text{safe}}$ for all $v_j \in \mathcal{V}$, $c_i \in \mathcal{R}$, then:*

(i) Every provisioned identity has a cosine margin of at least Δ below the operating collision boundary τ with respect to all enrolled gallery identities at provisioning time.

(ii) For a subsequently enrolled real identity r', any v_j satisfying $\cos(v_j, r') \in [\tau_{\text{safe}}, \tau)$ does not *cause an operational collision (since $\cos(v_j, r') < \tau$), but lies within the monitoring zone and may trigger reassessment under the provisioning policy. Only $\cos(v_j, r') \geq \tau$ constitutes a collision requiring revocation.*

(iii) The safety buffer requires more *perturbation than provisioning at τ: since $\tau_{\text{safe}} < \tau$ and $d\alpha^*(0,\tau)/d\tau = -1/(\tau^2\sqrt{1-\tau^2}) < 0$, it follows that $\alpha^*(0, \tau_{\text{safe}}) > \alpha^*(0,\tau)$. Concretely, $\tau_{\text{safe}} = 0.360$, $\tau = 0.391$: $\alpha^* \approx 2.59$ versus 2.35.*

Proof. Part (i). The provisioning constraint gives $\cos(v_j, c_i) < \tau_{\text{safe}} = \tau - \Delta < \tau$.

Part (ii). A collision under the operating system occurs iff $\cos(v_j, r') \geq \tau$; if $\cos(v_j, r') < \tau$, no collision occurs regardless of whether $\cos(v_j, r') \geq \tau_{\text{safe}}$.

Part (iii). Monotonicity follows from $d\alpha^*(0,\tau)/d\tau = -1/(\tau^2\sqrt{1-\tau^2}) < 0$, so $\tau_{\text{safe}} < \tau$ implies $\alpha^*(0, \tau_{\text{safe}}) > \alpha^*(0,\tau)$. □

Remark 3. *The safety buffer reduces the available GV capacity bound. Since $\mu(\cdot, \tilde{d})$ is* strictly decreasing *in τ:*

$$\tau_{safe} < \tau \implies \mu(\tau_{safe}, \tilde{d}) \geq \mu(\tau, \tilde{d}) \implies \frac{1}{\mu(\tau_{safe}, \tilde{d})} \leq \frac{1}{\mu(\tau, \tilde{d})}, \tag{37}$$

so $A_{GV}(\tau_{safe}) \leq A_{GV}(\tau)$. The capacity cost depends strongly on the choice of Δ: at τ_{safe}=0.360, τ=0.391 (Δ=0.031, one operating step): $A_{GV} \geq 1.81 \times 10^9 \approx 2^{30.8}$, giving headroom $\approx$ 1,810× over $|\mathcal{V}|$=10^6, a comfortable margin. By contrast, at τ_{safe}=0.319 (Δ=0.072, two operating steps below): $A_{GV} \geq 2.38 \times 10^7 \approx 2^{24.5}$, leaving only $\approx 24\times$ headroom over one million identities, a tight constraint that may limit provisioning at scale. Practitioners should choose Δ based on the expected enrollment growth rate.

C.7 Empirical Effective Capacity Estimator

Distinction from ambient-sphere capacity. $A_{\text{GV}}(\tilde{d}, \tau)$ is a packing lower bound on the ambient sphere $\mathbb{S}^{\tilde{d}-1}$ under the submanifold approximation (Sec. C.4). The empirical estimator below measures a distinct quantity:

$$A_{\text{eff}}(\tau) := \frac{1}{p_{\text{coll}}(\tau)}, \qquad p_{\text{coll}}(\tau) = \Pr\big[\cos(v_j, r') \geq \tau\big], \tag{38}$$

the inverse empirical collision probability for a real identity r' drawn from $\mathcal{M}_{\text{face}}$. Under the idealized uniform $\mathbb{S}^{\tilde{d}-1}$ model, $p_{\text{coll}} = \mu(\tau, \tilde{d})$ and $A_{\text{eff}} = A_{\text{GV}}$ numerically; in general they are different mathematical objects measuring different things, as discussed below.

Statistical Model. Let $v_j \in \mathcal{V}$ ($j = 1, \ldots, N$) be provisioned virtual identities and let $r'_l \in \mathcal{R}_{\text{test}}$ ($l = 1, \ldots, L$) be unseen real identities drawn independently from $\mathcal{M}_{\text{face}}$. Define the collision indicator $X_{jl} = \mathbf{1}[\cos(v_j, r'_l) \geq \tau]$, with $\Pr[X_{jl} = 1] = p_{\text{coll}}(\tau) = 1/A_{\text{eff}}(\tau)$ by the definition of A_{eff}. The total collision count $C = \sum_{j=1}^{N} \sum_{l=1}^{L} X_{jl}$ satisfies:

$$C \sim \text{Poisson}\left(\frac{NL}{A_{\text{eff}}}\right), \tag{39}$$

by the Poisson limit theorem for sums of rare independent events, where the individual event probability $p_{\text{coll}} = 1/A_{\text{eff}}$ is small; the expected count $\lambda = NL/A_{\text{eff}}$ need not be much smaller than one.

Maximum Likelihood Estimator. The log-likelihood of model (39) is:

$$\ell(A) = -\frac{NL}{A} + C \log \frac{NL}{A} - \log C!\,. \tag{40}$$

Setting $d\ell/dA = 0$ yields:

$$\hat{A}_{\text{eff}}^{\text{MLE}} = \frac{NL}{C} = \frac{|\mathcal{V}| \times |\mathcal{R}_{\text{test}}|}{C}. \tag{41}$$

Confidence Interval. An exact $(1 - \alpha_{\text{CI}})$ confidence interval for A_{eff} is obtained by inverting the exact Poisson confidence interval for C. With $\chi^2_{\nu,q}$ denoting the q-th quantile of the chi-squared distribution with ν degrees of freedom:

$$\left[\frac{NL}{\frac{1}{2}\chi^2_{2(C+1),\, 1-\alpha_{\text{CI}}/2}}, \; \frac{NL}{\frac{1}{2}\chi^2_{2C,\, \alpha_{\text{CI}}/2}} \right]. \tag{42}$$

This interval maintains the stated coverage probability under the Poisson model regardless of sample size.

Zero-Collision Lower Bound. When $C = 0$, the MLE (41) is undefined. From $P(C = 0) = e^{-NL/A_{\text{eff}}} \geq 0.05$:

$$\hat{A}_{\text{eff}} > \frac{NL}{\ln 20} \approx \frac{NL}{2.996}. \tag{43}$$

For $|\mathcal{V}| = 1\text{M}$, $|\mathcal{R}_{\text{test}}| = 180\text{K}$:

$$NL = 1.8 \times 10^{11}, \qquad \hat{A}_{\text{eff}} > 6.0 \times 10^{10} \approx 2^{35.8}. \tag{44}$$

$\hat{A}_{\text{eff}}$ and A_{GV} measure different quantities: $\hat{A}_{\text{eff}}^{-1}$ estimates the per-pair collision probability between provisioned virtual identities and held-out real identities under the empirical real-face distribution, while A_{GV} lower-bounds the ambient-sphere packing number under a uniform-measure model (Sec. C.4). They are complementary capacity references — empirical collision density and model-based combinatorial capacity — not competing bounds on the same object. The numerical relationship between them depends on how well the uniform-sphere model approximates the empirical real-face distribution; we make no claim that one validates the other.

Poisson Approximation Validity. The Poisson approximation requires individual collision probabilities $\mu(\tau, \tilde{d}) \ll 1$, not necessarily $\lambda = NL \cdot \mu \ll 1$. Across all six operating points, the cap probability remains small: the largest value occurs at the most permissive threshold τ=0.319, where $\mu = 4.21 \times 10^{-8}$; at the primary operating point τ=0.391, $\mu = 1.35 \times 10^{-11}$; and at the strictest recognition threshold τ=0.448, $\mu = 4.92 \times 10^{-15}$. These small individual collision probabilities support the Poisson approximation under the rare-event model. The expected total collision count λ varies substantially across operating points. For $|\mathcal{V}|$=1M and $|\mathcal{R}_{\text{test}}|$=180K, giving $NL = 1.8 \times 10^{11}$:

$$\lambda(0.319) = 1.8 \times 10^{11} \times 4.21 \times 10^{-8} \approx 7.58 \times 10^{3}, \tag{45}$$

$$\lambda(0.391) = 1.8 \times 10^{11} \times 1.35 \times 10^{-11} \approx 2.43, \tag{46}$$

$$\lambda(0.448) = 1.8 \times 10^{11} \times 4.92 \times 10^{-15} \approx 8.9 \times 10^{-4}. \tag{47}$$

At τ=0.319, $\lambda \gg 1$, so zero-collision bounds are not meaningful and the MLE Eq. (41) should be used directly. At τ=0.391, $\lambda \approx 2.43$: C=0 with probability $\approx 9\%$, C=1 with $\approx 21\%$, and $C \geq 2$ with $\approx 70\%$; the MLE applies in the majority of cases. At τ=0.448, $\lambda \ll 1$ and C=0 is near-certain ($> 99.9\%$), so the zero-collision bound Eq. (43) will typically apply. We verify stability of $\hat{A}_{\text{eff}}$ as $|\mathcal{R}_{\text{test}}|$ grows from 36K to 180K (Sec. 4); flat curves confirm that estimates are not dominated by local demographic density effects.

Assumption Audit. The estimator (41) rests on two assumptions:

(i) Uniform distribution on $\mathcal{M}_{\text{face}}$. Real identities cluster by demographic factors (ethnicity, age, gender), so r'_l is not strictly uniform over $\mathcal{M}_{\text{face}}$. If $\mathcal{V}$ resides in a demographically sparse region, $\hat{A}_{\text{eff}}$ is biased upward. The stability check described above detects this.

(ii) Independence of collision events. X_{jl} and $X_{j'l}$ for different j, j' at fixed l are not exactly independent when v_j and $v_{j'}$ are close. The inter-class separability constraint of Eq. (3) guarantees $\cos(v_j, v_{j'}) < \tau$ for all $j \neq j'$; under the Poisson limit theorem, the approximation in (39) holds under weak dependence when $p_{\text{coll}} \ll 1$, which is satisfied here.

C.8 Relationship between Capacity, α, and PCA Noise

The ambient-sphere reference $A_{\mathrm{GV}}(\tilde{d}, \tau)$ is determined by $(\tau, \tilde{d})$ alone and is independent of the proposal parameters α and κ. These parameters control how efficiently the algorithm proposes candidates that pass the hard BIP checks.

Define the valid region at step t:

$$\mathcal{V}_t^* = \left\{ v \in \mathbb{S}^{d-1} : \cos(v, c_i) < \tau \ \forall c_i \in \mathcal{R},\ \cos(v, v_j) < \tau \ \forall v_j \in \mathcal{V}_t \right\}, \tag{48}$$

and the acceptance probability at step t:

$$P_{\text{accept}}^{(t)}(\alpha, \kappa) = \Pr\Big[\text{normalize}(r + \alpha z) \in \mathcal{V}_t^*\Big]. \tag{49}$$

Then:

$$\mathbb{E}[|\mathcal{V}|] = \sum_{t=1}^{N_{\text{attempts}}} P_{\text{accept}}^{(t)}(\alpha, \kappa) \ \approx\ \bar{P}_{\text{accept}}(\alpha, \kappa) \cdot N_{\text{attempts}}, \qquad \mathbb{E}[|\mathcal{V}|] \le A_{\text{pack}}, \tag{50}$$

where $\bar{P}_{\text{accept}}$ is the average acceptance probability over the provisioning process.

$\kappa = 0$ **(pure repulsion):** $z = z^*$ provides directional bias away from the neighborhood centroid but yields low diversity across repeated candidates sampled from the same reference r.

$\kappa \to \infty$ **(pure PCA noise):** Under a crude disjoint-cap and uniform-density approximation:

$$P_{\text{accept}}^{(t)} \approx 1 - (M + |\mathcal{V}_t|)\, \mu(\tau, \tilde{d}) = 1 - \frac{M + |\mathcal{V}_t|}{A_{\mathrm{GV}}}. \tag{51}$$

At $\tau = 0.391$ and $|\mathcal{V}| = 10^6$:

$$1 - \frac{360{,}232 + 10^6}{7.41 \times 10^{10}} \approx 99.998\%. \tag{52}$$

For the 10M embedding-allocation experiment:

$$1 - \frac{360{,}232 + 10^7}{7.41 \times 10^{10}} \approx 99.986\%. \tag{53}$$

The acceptance probability remains near unity throughout provisioning at both scales.

PCA-aware weighting by σ_k *biases* perturbations toward high-variance identity directions and suppresses low-variance directions, keeping candidates close to the empirically dominant face-identity subspace. The sum in Eq. (6) runs over all $d = 512$ components, so perturbations are not strictly confined to the top-$\tilde{d}$ subspace, but the contribution from low-variance directions is substantially down-weighted by σ_k, although the residual variance beyond the top-$\tilde{d}$ components is not exactly zero.

D Additional Implementation Details of G (GapGen)

Fig. 8 summarizes the GapGen training pipeline. Below we describe the base pipeline, conditioning, sampling, round-trip re-encoding, and the gap-aware fine-tuning curriculum in turn.

Base pipeline. The SDXL UNet and text encoders are frozen; the two identity-aware components, the IP-Adapter cross-attention projections and the IdentityNet, are jointly fine-tuned.

Conditioning. Because BIP allocates and renders identities in the same ArcFace embedding space, a provisioned $s \in \mathcal{V} \subset \mathbb{R}^{512}$ is fed directly through the pipeline's `image_embeds` input *without* any face image, face detector, or projection MLP on the condition side. We use a fixed canonical 5-point keypoint layout (face scale 0.28 of the canvas, y-offset -5%) as the IdentityNet condition for every identity.

Sampling. All images are generated at 1024×1024 with steps=30, guidance_scale=3.0, controlnet_scale=0.8, ip_adapter_scale=0.8, and the default SDXL scheduler. For virtual step, we use a single fixed prompt (*"candid color portrait photo of a person, natural lighting"*) and negative prompt suppressing CGI/cartoon styles, watermarks, oversmooth skin, and low-resolution artifacts.

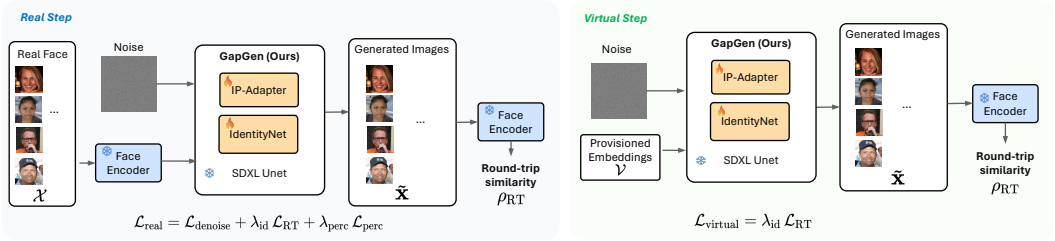


Figure 8: GapGen training pipeline. Real and virtual steps are interleaved at mixing ratio β during fine-tuning. **Real step** (left): real face $x \in \mathcal{X}$ is encoded by the frozen face encoder ϕ to produce target $e^* = \phi(x)$; GapGen renders $\tilde{x} = G(e^*)$, supervised by $\mathcal{L}_{\mathrm{denoise}} + \lambda_{\mathrm{id}}\mathcal{L}_{\mathrm{RT}} + \lambda_{\mathrm{perc}}\mathcal{L}_{\mathrm{perc}}$. **Virtual step** (right): a provisioned embedding $s \in \mathcal{V}$ is fed directly as condition with no paired ground-truth image; supervision relies solely on the round-trip identity loss $\mathcal{L}_{\mathrm{RT}} = 1 - \cos(\phi(\tilde{x}), s)$. IP-Adapter cross-attention projections and IdentityNet are jointly fine-tuned; SDXL UNet and the face encoder remain frozen throughout.

Round-trip re-encoding. We re-extract embeddings from $\tilde{x} = G(s)$ with the same frozen ArcFace pipeline used during allocation: `antelopev2` face detection followed by `glintr100` (IResNet-100). The re-encoded $\tilde{e} = \phi(\tilde{x})$ is directly comparable to the target s via cosine similarity since both lie on the same 512-d unit sphere.

Fine-tuning. We initialize the IP-Adapter from the pretrained InstantID checkpoint and IdentityNet from the pretrained ControlNet, then jointly fine-tune both on a gap-aware curriculum of $(s, \tilde{x})$ pairs (Sec. 3.3) with the standard denoising loss. SDXL UNet and text encoders remain frozen throughout.

E Verification, Visual Examples, and IAPCT

E.1 v-LFW Protocol and Visual Examples

Construction. We construct **v-LFW** as a structural counterpart to LFW: it inherits the identity count and the official 10-fold $\times$ 600-pair split, but populates every identity slot with a freshly allocated virtual identity $s_n \in \mathcal{V}$ (distinct from any real LFW person by BIP non-collision), rendered as $\tilde{x}_n = G(s_n)$ at 1024×1024 following the conditioning in App. D.

Protocols. The shared pair list is then reused under four protocol families:

- **R-R**: standard LFW verification on original real images. Each pair (x_a, x_b) from LFW is scored by $\cos(\phi(x_a), \phi(x_b))$ and thresholded at the per-fold operating point τ_{ver}.
- **V-V**: same protocol as R-R but on v-LFW images at the matched slot positions, so the same/different structure is preserved over virtual identities.
- **R-V**: one image taken from LFW, the other from v-LFW at the same slot index. Every R-V pair crosses both identity and reality by construction, so its score distribution is a non-mate distribution. We report FAR at the threshold calibrated on R-R same-identity scores at TAR = 95%.
- **Detection**: per-image binary classification of real vs. virtual without identity labels, evaluated on the union of LFW and v-LFW images. We report AUC of the IAPCT forensics score $\hat{p}(x)$ (Sec. E.2).
- **Unified**: joint recognition and detection on the combined LFW + v-LFW pair set. A pair is unified-correct iff verification and the per-image real-vs-virtual decisions are all correct; the formal indicator is given in Eq. (56) below.

Visual examples.

Fig. 9 shows multiple images rendered from the same virtual identity $s \in \mathcal{V}$ under varying pose configurations, lighting conditions, and text style prompts. Each row corresponds to one virtual identity; columns show three independent generations sharing the same conditioning embedding s but differing in keypoint layout, prompt, and sampling seed. Identity is preserved across columns within each row, while pose, expression, lighting, and background vary, supporting the within-identity variation required for v-LFW (Tab. 3b).



Figure 9: v-LFW visual examples. Each row shows three images rendered from a single virtual identity $s \in \mathcal{V}$ at α=4, varying pose, lighting, and style while preserving identity.

E.2 IAPCT: Identity-Anchored Patch Consistency Transformer

Design. IAPCT augments the frozen ArcFace backbone with a transformer head that interrogates whether local patch statistics at each intermediate layer are consistent with the global identity embedding $e = \phi(x)$. The recognition path is preserved verbatim; the forensics path is additive and introduces no overhead at inference beyond the transformer forward pass.

Backbone and intermediate features. We use IResNet-100 with pretrained ArcFace weights. For an aligned input $x \in \mathbb{R}^{3\times112\times112}$, spatial feature maps are tapped at four intermediate stages: $\{F_l\}_{l=1}^{4}$, with spatial resolutions 56×56, 28×28, 14×14, 7×7 and channel widths 64, 128, 256, 512 respectively. $e_5 = \phi(x) \in \mathbb{R}^{512}$ is the standard L2-normalized ArcFace identity output, left unchanged for downstream recognition.

Tokenisation. For each stage l, all spatial positions p of F_l are extracted as patch tokens and projected to d_{model}=256 via a per-stage linear layer:

$$t_{l,p} = W_l\, F_l[:, p] \in \mathbb{R}^{256}, \quad p = 1, \ldots, H_l W_l. \tag{54}$$

The identity anchor is projected as: $e_{\text{proj}} = W_{\text{id}}\, e \in \mathbb{R}^{256}$.

Transformer encoder. A 4-layer transformer encoder processes the patch tokens $\{t_{l,p}\}$ from all four stages jointly. Each layer applies self-attention among spatial tokens followed by cross-attention to the identity anchor e_{proj}, producing patch-identity consistency scores:

$$\gamma_{l,p} = \sigma\left(\frac{(W_Q\, t_{l,p}) \cdot (W_K\, e_{\text{proj}})}{\sqrt{d_k}}\right) \in (0, 1), \tag{55}$$

where σ is the sigmoid function. Real faces yield uniformly high $\gamma_{l,p}$ since every spatial region originates from the same physical person; virtual faces yield heterogeneous $\gamma_{l,p}$ due to unconstrained intermediate features.

Head and loss. A `[CLS]` token produces a forensics logit through:

$$\text{LN}(256) \to \text{Linear}(256{\to}128) \to \text{GELU} \to \text{Dropout}(0.1) \to \text{Linear}(128{\to}1).$$

Training loss: $\mathcal{L} = \text{BCE}(\hat{y}, y) + \lambda_c(H(\gamma^{\text{virtual}}) - H(\gamma^{\text{real}}))$, where $H(\gamma^{(x)}) = -\sum_{l,p} \bar{\gamma}_{l,p}^{(x)} \log \bar{\gamma}_{l,p}^{(x)}$ is the entropy of the normalized attention distribution, y=0 for real and y=1 for virtual, and ϕ remains frozen throughout.

Training. Real samples: Glint360K 112×112 aligned crops. Virtual samples: 100K images at α=4 from GapGen. Batch size 128 with 30% virtual fraction; weighted random sampling. AdamW, lr 10^{-4}, weight decay 10^{-2}, gradient clip 5.

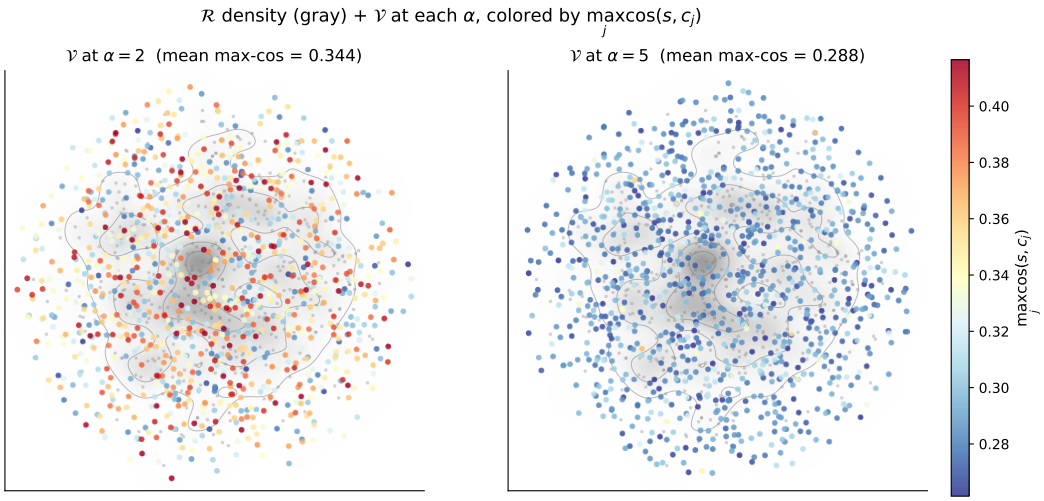


Figure 10: t-SNE of real and virtual identities. Real centroids $\mathcal{R}$ are shown as a gray density background. Each virtual identity in $\mathcal{V}$ is coloured by $\max_{c_j \in \mathcal{R}} \cos(s, c_j)$ (red: close to a real cluster; blue: located in a low-density gap). Increasing α shifts virtual identities from regions dense with real centroids (α=2, mean max-cos 0.344) toward low-density gaps (α=5, mean max-cos 0.288). No internal collapse is observed within $\mathcal{V}$.

Unified metric. A pair (x_a, x_b) is unified-correct iff verification and the two per-image real-vs-virtual decisions are all correct:

$$\mathbf{1}_{\text{uni}} = \mathbf{1}[\hat{y}_{\text{ver}}{=}y_{\text{ver}}] \cdot \mathbf{1}[\hat{y}^a_{\text{rv}}{=}y^a_{\text{rv}}] \cdot \mathbf{1}[\hat{y}^b_{\text{rv}}{=}y^b_{\text{rv}}], \tag{56}$$

with $\hat{y}_{\text{ver}} = \mathbf{1}[\cos(e_5(x_a), e_5(x_b)) \geq \tau_{\text{ver}}]$ at the per-fold LFW threshold and $\hat{y}^{\bullet}_{\text{rv}} = \mathbf{1}[\hat{p}(x_{\bullet}) \geq 0.5]$. Unified accuracy averages $\mathbf{1}_{\text{uni}}$ over the union of R-R, V-V, and R-V pairs. The joint indicator (rather than an average over verification and detection accuracies) matches the deployment scenario where a system must simultaneously verify the identity *and* flag the face as real or virtual.

F Additional Results

F.1 t-SNE of $\mathcal{R}$ vs. $\mathcal{V}$

Fig. 10 shows a t-SNE projection of real identity centroids $\mathcal{R}$ (rendered as a gray density background) together with virtual identities $\mathcal{V}$ at $\alpha \in \{2, 5\}$ (1K samples per panel). Each point in $\mathcal{V}$ is coloured by $\max_{c_j \in \mathcal{R}} \cos(s, c_j)$, i.e., its cosine similarity to the nearest real centroid: red indicates virtual identities that lie close to some real cluster, while blue indicates identities that fall into low-density gaps between clusters. Three observations are consistent with Obs. 1. (i) $\mathcal{V}$ interleaves with $\mathcal{R}$ rather than forming a separate cloud: virtual identities appear throughout the manifold, including regions dense with real centroids. (ii) Increasing α from 2 to 5 systematically pushes $\mathcal{V}$ away from real clusters and into the gaps: the mean max-cos decreases from 0.344 (α=2) to 0.288 (α=5), consistent with the non-collision improvement reported in Tab. 1.

F.2 Additional Image Grids

Fig. 11 shows 40 randomly selected faces generated by G from virtual identities $s \in \mathcal{V}$ at α=4. The samples exhibit substantial diversity in gender, age, ethnicity, hairstyle, and accessories (e.g. glasses), while showing no obvious identity duplication across images. This provides qualitative evidence that the perturbation-based provisioning scheme, $s = r + \alpha z$, yields novel and diverse identities without collapse, even at larger scale.



Figure 11: Sample virtual identities from BIP. Forty randomly sampled virtual identities rendered by GapGen at 1024×1024, spanning diverse demographics, age, and appearance.